Network transparency for a local LLM: Qwen3-14B v2 NT

A LoRA fine-tune that announces and reports every network connection — results, statistics and an honest account of two shortfalls

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4 November 2026 — preprint, version 1.0

Abstract

We fine-tuned **Qwen3-14B v2** (Qwen/Qwen3-14B, Apache-2.0, with the author's earlier self-distillation identity LoRA merged) with a LoRA adapter so that, whenever it uses network tools, the model **announces before** and **reports after** every connection: host, country and direction. The selected full fine-tune (**`qwen14-scienziato-v2-nt`**, adapter scale 1.0) reaches 99.7% connection recall, 100% country and direction accuracy and 99.3% pre-announcement on 450 development runs; GSM8K and ARC-it are statistically unchanged and identity improves significantly (98% vs 76% on a sealed test). **Two pre-registered criteria were not met** and are disclosed up front (Section 6): IFEval is 0.4 points lower than the base (83.5% vs 83.9%, not significant), and the unrequested-network-attempt ("leak") rate is 1.11% (5/450) versus a pre-registered gate of ≤1%, with 95% Wilson CI [0.48%, 2.57%]. The model is published as a documented exception to the author's "beats base without regressions" rule.

1. Motivation

Local LLMs increasingly call tools that touch the network. A user should know *which* connections happen, *where* they go and *in which direction*, before and after they happen. We train this behaviour into the model and measure it against an independent packet-level ground truth.

2. Method

- **Base**: Qwen/Qwen3-14B (revision 40c06982) + merged identity LoRA `odelpa/qwen14-scienziato-v2` ("base v2").
- **Adapter**: LoRA r = 16, α = 16, dropout 0.05, on q, k, v, o, gate, up, down projections.
- **Data**: synthetic tool-use dialogues with real/mock network calls plus a replay mix of general data to prevent forgetting; 90 development cases × 5 repetitions (T = 0.7) = 450 runs; separate sealed test.
- **Training**: 2 epochs, 786 steps, one rented H100 NVL GPU (~57 min of training); validation loss NT 0.067, replay 0.118.
- **Candidates**: the adapter at scales s ∈ {0.3, 0.5, 0.7, 0.85, 1.0}; selection by pre-registered gates on the dev set.

Metrics

With C the set of real connections (from the packet capture) and $\hat{C}$ the set the model reports:

$$\text{recall} = \frac{|C \cap \hat{C}|}{|C|}, \qquad \text{precision} = \frac{|C \cap \hat{C}|}{|\hat{C}|}, \qquad \text{leak} = \frac{\#\{\text{runs with unrequested network}\}}{\#\{\text{runs where no network was requested}\}}.$$

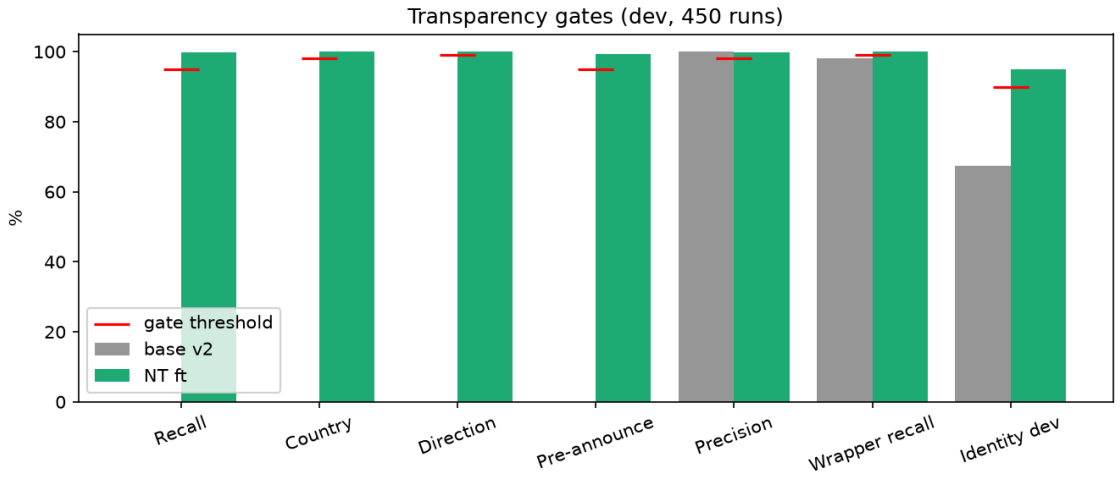
Proportions are reported with 95% Wilson intervals:

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \qquad z = 1.96.$$

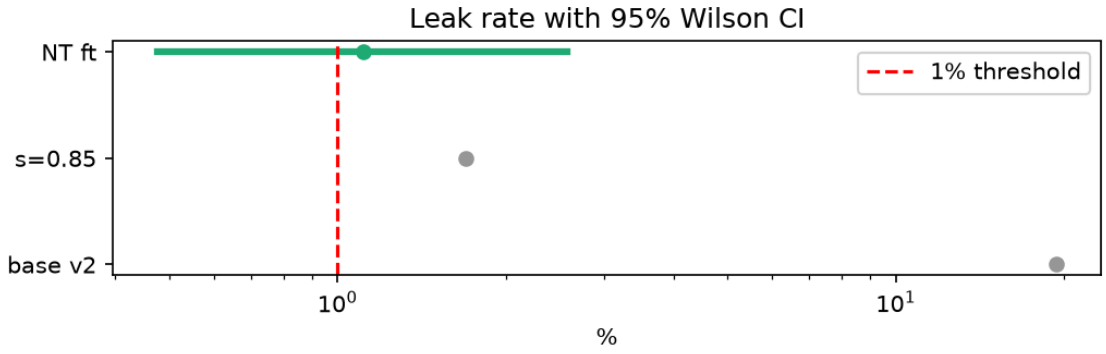
Paired benchmark differences use the exact McNemar test on discordant pairs (b, c): $p = 2\sum_{k=0}^{\min(b,c)} \binom{b+c}{k} 2^{-(b+c)}$.

3. Transparency gates

Gate	Threshold	Base v2	NT ft	Result
Connection recall	≥ 95%	0.0	99.7	✓
Country	≥ 98%	0.0	100	✓
Direction	≥ 99%	0.0	100	✓
Pre-announcement	≥ 95%	0.0	99.3	✓
Precision (no phantom connections)	≥ 98%	100	99.8	✓
Leak attempt	**≤ 1%**	19.4	**1.11**	**✗ missed (+0.11 pt)**
False refusal	≤ 3%	6.7	0.4	✓
Wrapper recall	≥ 99%	98.1	100	✓
Wrapper leak executed	= 0	0	0	✓
Identity (dev)	≥ 90%	67.5	95.0	✓



Transparency gates

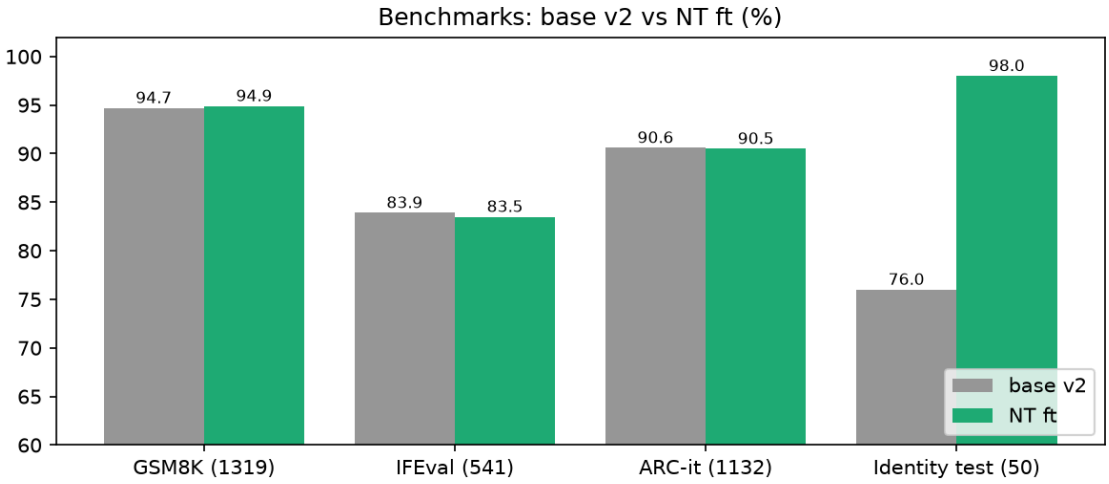


Leak rate with Wilson CI

Candidate comparison: s = 0.85 had leak 1.7% and identity 92.5%; s = 0.7 failed direction and pre-announcement. The full-scale adapter is the best on both leak and identity.

4. Capability benchmarks (base v2 vs NT ft)

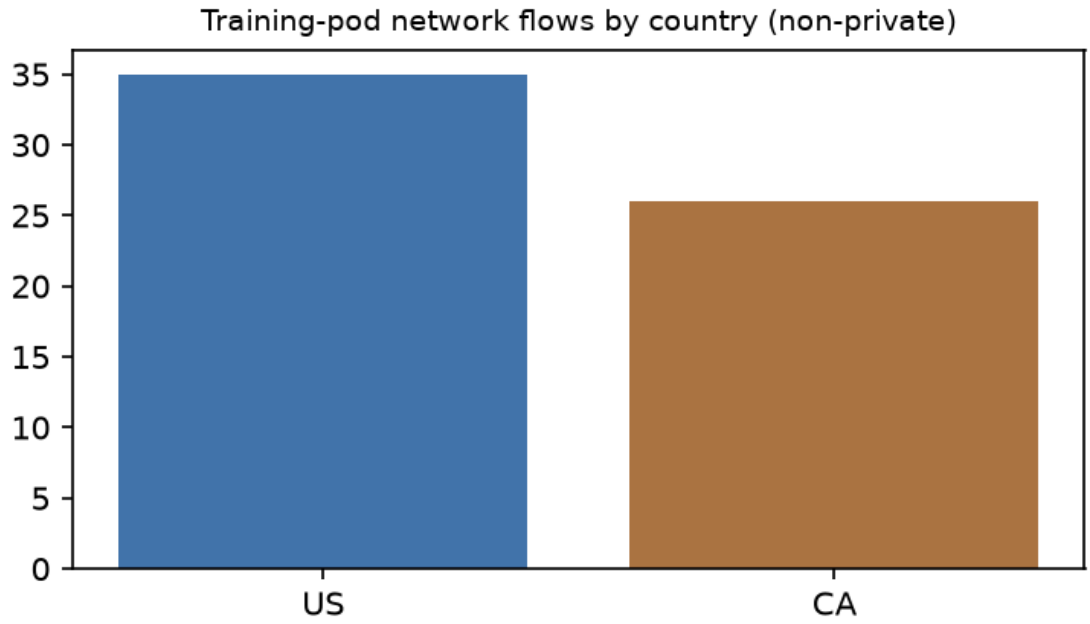
Set	n	Base v2	NT ft	Δ (pt)	McNemar p
GSM8K	1319	94.7	94.9	+0.2	0.70
IFEval	541	83.9	**83.5**	**−0.4**	0.87
IFEval held-out subset	141	82.3	80.8	−1.4	0.73
ARC-it	1132	90.6	90.5	−0.2	0.63
Identity (sealed test)	50	76.0	98.0	+22.0	0.001
Identity (dev)	40	67.5	95.0	+27.5	0.003



Benchmarks

5. Network audit of the training run

The training machine itself was monitored with a packet sniffer (≈6.2 M packets) and offline GeoIP (DB-IP Lite): 63 flows, **0 anomalies**; non-private destinations US 35, CA 26, all attributable to package mirrors (PyPI, PyTorch), Hugging Face downloads and the operator's SSH sessions. Total compute cost of the final run: US$ 8.48 (whole NT project ≈ US$ 13.84).



Network flows

6. Limitations (read before use)

1. **IFEval regression.** The fine-tune scores 83.5% vs 83.9% for the base (−0.4 pt, McNemar p = 0.87; −1.4 pt on the 141-item held-out subset, p = 0.73). The difference is not statistically significant, but the point estimate is lower, so the strict "no regressions" criterion is **not** met.
2. **Leak gate missed.** Unrequested network attempts occur in 5/450 dev runs = 1.11%, above the pre-registered ≤1% gate. 95% Wilson CI [0.48%, 2.57%]: the threshold lies inside the interval, so the miss is not statistically distinguishable from the gate — but the upper bound means the true rate could be up to ~2.6%. **Always run the model behind a network wrapper/firewall**; the transparency is a reporting aid, not a security boundary.
3. Evaluation uses a mock tool environment plus limited real-network windows; the leak rate should be re-measured on a larger sealed test.
4. The adapter is trained on top of *base v2* (Qwen3-14B + the identity LoRA), not on vanilla Qwen3-14B.
5. Acceptance was a documented gate override by the author (4 Nov 2026), not an automatic pass.

7. License and attribution

Base model Qwen3-14B © Alibaba Cloud / Qwen team, released under the **Apache License 2.0**, which permits commercial use, modification and redistribution provided the license text is included, notices are kept and modifications are stated. This adapter is a modification of Qwen3-14B and is distributed under Apache-2.0.

Citation

Del Papa, G. (2026). *Network transparency for a local LLM: Qwen3-14B v2 NT.* Lo Scienziato Dilettante GDP. DOI: to be assigned by Zenodo.